PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: SupplyLah

Version: 1.0

1. Project Overview

- **Problem Statement:** Small and medium-sized enterprises (SMEs) in Malaysia's wholesale and distribution sector rely heavily on chat platforms like WhatsApp to manage supply chain orders, but these messages are unstructured, multilingual and require manual processing. The high volume of text, voice messages and calls received daily requires significant human effort to process orders, manage inventory, and coordinate logistics, resulting in inefficiencies and errors.
- **Target Domain:** Enabling agentic AI systems for end-to-end automation of supply chain workflows in SME wholesale and distribution sectors.
- **Proposed Solution Summary:** SupplyLah removes the hassle of manual supply chain operations by converting unstructured chat, voice messages and voice calls from customers placing wholesale customers sent through messaging platforms into detailed and automated workflows using a multi-agent AI system that handles complex order processing, inventory decisions, and logistics coordination.

2. Background & Business Objective

- **Background of the Problem:** For decades, the wholesale supply chain in Malaysia has been operating on a highly fragmented, manual basis through social messaging apps like WhatsApp to manage daily orders. A single wholesaler often manages hundreds of daily orders sent as unstructured text, rapid-fire voice notes in Malaysian English, Malay and "bahasa rojak" which staff must manually read, cross-reference against physical whiteboards or Excel sheets for stock, calculate complex pricing, and manually book third-party logistics like Lalamove. This workflow is prone to human error, results in

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missed orders during festive peak seasons, and prevents businesses from scaling due to the sheer cognitive load of manual coordination.

- **Importance of Solving This Issue:** Automating supply-chain order intake and fulfillment helps SMEs overcome language and format barriers, allowing them to handle larger order volumes. It also overcomes language and format issues, allowing traditional merchants to accept multimodal inputs such as voice or images while maintaining a perfectly structured digital inventory. This reduces over-ordering, minimizes food and perishable-goods waste, and enables more stable pricing and stock rotation through better real-time inventory awareness. Such systems also reduce employee burnout and cognitive load, contributing directly to SDG 8 Decent Work and Economic Growth by making manual supply-chain roles more sustainable and to SDG 12 Responsible Consumption and Production by minimizing inventory drift and waste.

- **Strategic Fit / Impact:** It aligns perfectly with Ant International's mission to accelerate SME digitalization in Malaysia and Southeast Asia through platforms like Alipay+ and EPOS360 by bridging the gap between informal conversational commerce and formal fintech or logistics. Also, using Z.AI's GLM as the central reasoning engine allows the system to handle the ambiguity of Malaysian slang and mixed-language inputs, while remaining fully compliant with the hackathon's requirement that GLM is the core decision-making component.

3. Product Purpose

3.1. Main Goal of the System: To provide an autonomous stateful AI agent that manages the end-to-end wholesale lifecycle by transforming unstructured conversation into structured, actionable logistics and payment events.

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3.2. Intended Users (Target Audience):

- Wholesale Suppliers
- SME Distributors
- Retail Buyers
- Warehouse Managers

4. System Functionalities

4.1. Description: The system operates as an intelligent agentic supply chain automation engine that helps to transform unstructured wholesale customer orders from chat and voice messages into structured, executable workflows through multi-agent orchestration across order processing, inventory management, and logistics execution.

4.2. Key Functionalities:

- **Multi-modal Order Ingestion:** Transforms unstructured wholesale customer inputs from chat and voice messages into structured order data, extracting key entities such as products, quantities, customer details, and delivery location.

- **Voice Call Order Processing**: Captures voice call transcripts between sellers and customers, identifying order intents to autonomously trigger downstream inventory updates and logistics scheduling.

- **Adaptive Inventory Orchestration:** Real-time stock availability checking and generating intelligent substitution or pricing recommendations when inventory is insufficient.

- **Logistics Automation & Coordination:** Delivery platforms integration, real-time price fetching and automated booking once the order is finalized.

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- **Conversational Customer Updates:** Provides automated responses and updates to customers through messaging platforms like order confirmations, stock updates, substitutions, and delivery status notifications.

- **Autonomous Failure & Ambiguity Handling:** Detects incomplete or ambiguous customer orders, such as missing quantities, unclear product types, or vague instructions, and resolves them by prompting for clarification or leveraging past order history to suggest likely items for confirmation.

- **Operational Dashboard & Monitoring**: Provides a centralized interface for wholesalers to monitor incoming orders, track order status, view inventory levels, and oversee logistics operations in real time.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Our system leverages three different GLM models to optimize decision-making while maintaining cost efficiency for small businesses. First, we use ILMU-GLM-5.1 as the main GLM model for our most complex tasks like understanding messy phone call transcripts, ambiguous order requests and stock substitutions due to its strong reasoning capabilities for long-horizon autonomous tasks. For image recognition and OCR extraction of handwritten or printed order lists, GLM-4.6V is chosen due to its native support for multimodality and direct processing of documents containing visual elements such as tables and images. We used Groq API to access OpenAI Whisper Large V3 for audio input processing in voice messages and voice call due to its excellent at turning speech into text, especially in multilingual and dialect processing, that are common in real-world wholesale customer interactions.

4.3.2. Prompting Strategy: Our system uses multi-step agentic prompting strategy, where each agent is assigned a specific task with structured prompts tailored to its role. The intake specialist agent uses structured prompting, enhanced with few-shot prompting containing highly realistic examples of mixed-language phrases to extract key entities such as items,

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quantities, and delivery details from unstructured, multilingual inputs. Additionally, this agent employs reasoning to verify if a minimum viable order is met; if confidence is low or details are missing, it dynamically drafts a clarification message in the buyer's exact language. The logic and inventory orchestrator agent applies guided prompts with business rules to perform inventory checks, substitution decisions, and pricing logic. The logistics and state manager agent uses task-specific prompts to coordinate delivery scheduling and status updates. This approach is chosen because this supply chain workflow involves multiple dependent decisions that cannot be solved in a single prompt. By breaking the workload into multi-step agentic roles, the system enables more accurate reasoning and consistent supply chain decision-making.

4.3.3. Context & Input Handling: The system is designed to handle unstructured and variable-length inputs from customers, including text messages, voice notes or voice calls from customers. For text inputs, inputs that exceed the limit will be automatically segmented into smaller chunks and will be merged together again later before being passed to the intake agent. For audio inputs, voice messages or voice calls will be processed into segments during transcription to ensure accurate speech-to-text conversion in multilingual and dialect speeches. Background noises of audio messages and the voice call record will be removed using noise reduction techniques during preprocessing. For audio messages that are unclear, excessively noisy and can’t be identified by their context, the system will be flagged for clarification and request a refined response from the user.

4.3.4. Fallback & Failure Behavior: The system is designed for multiple fallback mechanisms when the model produces off-topic and hallucinated outputs. Each agent will validate the output before proceeding and passing to the next stage of the workflow. For instance, there will be a confidence level checking for the output given by cross-referencing the output with the business rules. If the confidence level is low, the system triggers a retry with refined prompting to generate a more accurate response. If repeated attempts still fail, the system will isolate the affected request and prompt the customer for clarification and confirmation of order details. Critical failures are flagged and queued for human review through the operational dashboard.

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5. User Stories & Use Cases

- **User Stories:**

  "As a wholesaler, I want an AI to read my 200+ WhatsApp messages so that I don't miss out orders or sell stock I don't actually have during festive peak seasons."

  "As an SME Distributor, I want to automatically calculate delivery fees and book third-party logistics like Lalamove as soon as an order is confirmed, so that I can eliminate manual booking errors and ensure faster, more reliable deliveries to my clients."

  "As a Retail Buyer, I want to send a quick voice note or text in my native language and receive instant order confirmation or substitution offers if stock is low, so that I can finish my supply orders in seconds and get back to running my kitchen."

  "As a Warehouse Manager, I want the AI to cross-reference every incoming order against our real-time stock levels before it is confirmed, so that I can prevent overselling and maintain a 100% accurate view of my inventory on the dashboard."

- **Use Cases (Main Interactions):**

  Buyer Placing Order via Voice Note: Buyer sends a messy voice note via WhatsApp to place an order. The system immediately transcribes the audio, extracts the order details, checks the stock, and replies on WhatsApp with a structured order summary and total price for the wholesaler to confirm.

  Buyer Placing Order via Live Phone Call: Buyer calls the wholesaler directly to place their order. The wholesaler accepts the call through the system, and the system captures the live transcript and identifies the order intent. When the call ends, the AI analyzes the conversation to confirm the transaction intent and uses the transcript to instantly generate an official invoice and delivery quote, sending it to the buyer via WhatsApp for payment. Once the buyer completes the payment, the system automatically deducts the inventory and updates the order status to Dispatched on the wholesaler's dashboard.

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Buyer Negotiating Out-of-Stock Items: When the buyer orders an item A that is currently out of stock, the AI replies on WhatsApp, suggesting a substitute item B based on the seller's business rules. If the buyer agrees and sends a confirmation message, the system remembers the context, applies any rule-based discount, and generates the inventory and prepares the delivery order for the wholesaler.

Wholesaler Setting Up Business Rules: Wholesaler logs into the dashboard for the first time and uploads an Excel document of their inventory, inputs their customer alias list, and types in their custom discount rules. The AI will read the Excel document, digitize the inventory into a Supabase and train the agents on the written pricing logic so it can start taking orders autonomously.

AI Clarification for Ambiguous Order: If a buyer sends an extremely confusing message for a product that does not exist, the AI Agent will check the database and automatically reply to the buyer to ask for clarification. If the buyer replies with something the AI still cannot process, the AI safely pauses the workflow and flags the order on the seller's Operational Dashboard with a red alert.

Real-Time Operational Oversight and Monitoring: The wholesaler admin can log into the main dashboard to get a bird's-eye view of all operations without needing to interact with the AI agents directly. The system provides a real-time Command Center view, including Live Order Kanban, Inventory Status, Logistics Tracker, and AI Activity Log.

Resolving AI Exception Alerts: Wholesaler admin logs into the dashboard and sees the high-priority alert card. Upon clicking, the dashboard will show the raw output by the agent of the buyer alongside the reason on why it failed. The wholesaler admin manually overrides the order and completes it. The system then clears the alert and resumes the automated workflow.

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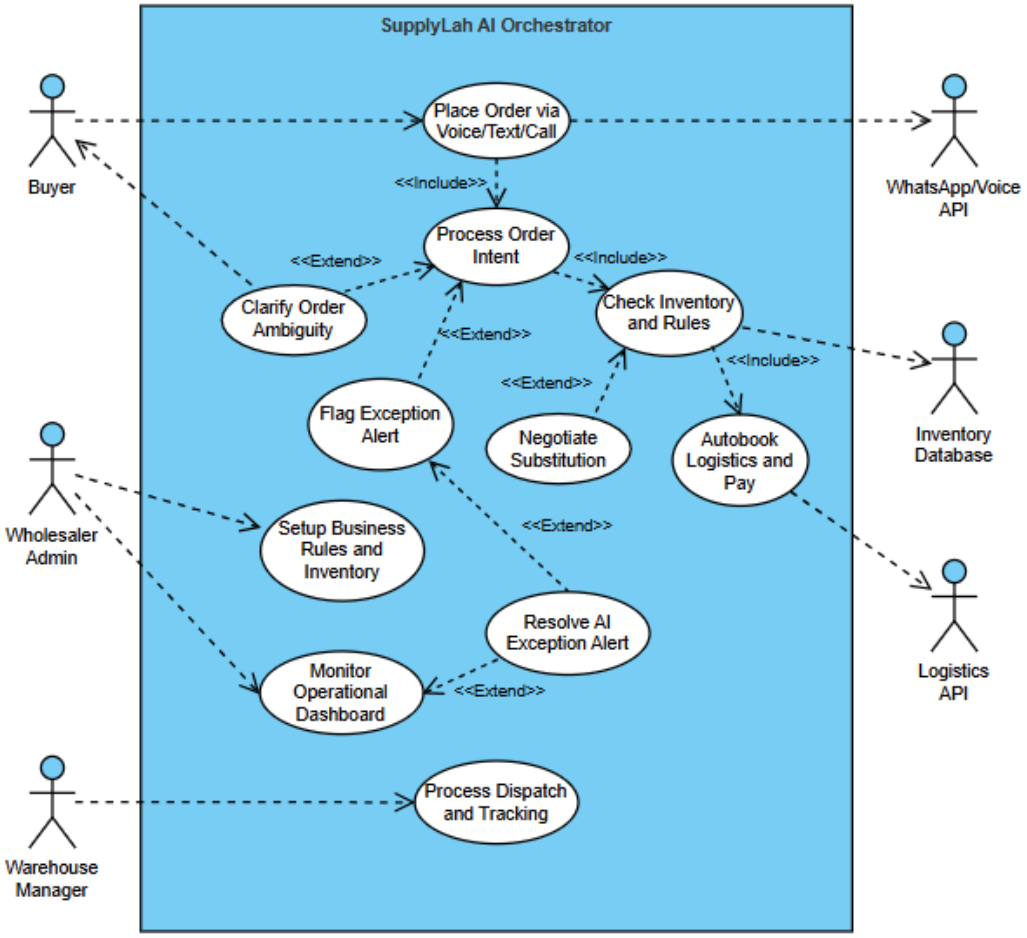
Warehouse Manager Fulfilling Confirmed Orders: The warehouse manager monitors the command center dashboard to see a live feed of fully processed and confirmed orders. They can view the structured list of items that need to be packed, confirm that the AI has automatically updated the inventory stock levels, and check the arrival time of the auto-booked Lalamove driver so they can prepare the goods for dispatch.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- **WhatsApp Message Intake**

  SupplyLah connects to the WhatsApp Business API via Twilio or other APIs, to receive inbound messages from clients and buyers on the wholesaler's existing business number. The system will monitor the inbox in real-time, routing each incoming message, be it plain text and photo of handwritten order list to the intake agent for processing. Buyers will experience no change to their current ordering behaviour by continuing to message the same number as usual.

- **Voice Notes Transcription to Order**

  Some buyers and clients will prefer sending voice notes instead of typed messages, particularly when involved complex order structures and multi-item requests. SupplyLah will automatically download the voice note attachments from WhatsApp and pass through OpenAI Whisper to produce a text transcript, which will be analysed by the GLM-5.1 from the intake agent together with the message metadata. The system will be mainly handling Malaysian English and Bahasa Melayu, where the documented limitation will be heavy regional accents and improper spoken language (Bahasa Rojak).

- **Live Voice Call Transcription to Order**

  For buyers who prefer to phone in for their orders, SupplyLah will intercept the inbound WhatsApp voice calls to business numbers and route the audio through a speech-to-text pipeline via OpenAI Whisper. The transcribed text will be passed to the intake agent for order parsing, similar to the same GLM-5.1 pipeline used for voice notes and text messages. Latency between speech and transcript generation is one of the known constraints, where the system will process the order once the call ends rather than attempting mid-call parsing.

- **Handwritten Order Image Parsing**

  Buyers and clients who take photographs of handwritten order lists and send them as images via WhatsApp are supported through GLM's multimodal vision capabilities. The

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intake agent will pass the image directly to Z.ai GLM-4.6V, which interprets the handwritten content and performs Optical Character Recognition (OCR). Then, GLM-5.1 extracts product names, quantities etc. in a structured JSON format as text and voice orders.

- **GLM-Powered Multilingual Order Parsing**

  Using ILMU-GLM-5.1 for core reasoning engine, the intake agent can parse unstructured, multilingual messages and ambiguous references in the orders, where it will output a structured JSON format object. If a minimum viable order cannot be extracted, the system will compose a clarification message in the buyer's language before proceeding with the order, and allow the system to send the message via the WhatsApp business account.

- **Live Inventory Read from Supabase**

  The inventory and business logic agent will get the wholesaler's current stock levels and business rules via Supabase at the time when each order is being processed. With Supabase PostgreSQL implementation, we can allow for database row-level locking mechanism, preventing concurrency issues and ensuring atomicity when deducting stock levels.

- **GLM-Powered Inventory Reasoning**

  When the requested stock is partially or fully unavailable, the inventory and business logic agent will invoke ILMU-GLM-5.1 for an appropriate business response. GLM will consider the stock deficit and available substitute products, to generate a substitute proposal, together with the business rules. This includes recalculating the revised order total if any discounts are applied and composing the message in natural language. For this MVP, it will only support a single-level substitution mechanism.

- **Customer Confirmation Loop**

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After sending the pricing quotes or substitution proposal, the system will enter a stateful waiting state together with the order session. If the buyer replies with affirmative keywords within the next 30 minutes, the order will be confirmed and passed to the next agent for logistics. If no reply is received, a follow-up nudge is sent and the system waits for another 30 minutes. Lastly, if the order is yet to be confirmed, the order is marked as expired and logged in the dashboard. The timeout window and follow-up message will be configurable based on the wholesaler operator settings.

- **Inventory Deduction on Order Confirmation**

  After the buyer has confirmed on the order, the system will write and update the confirmed quantities to the Supabase, reducing available stock for each item before logistics booking is initiated. This prevents overselling when multiple orders are confirmed in rapid succession.

- **Logistics Booking via External API**

  After the order has been confirmed, the logistics agent will call Lalamove API (or GrabExpress API) to request a delivery quote, passing pickup address, delivery address and estimated cargo weight. The API response, which includes the vehicle type, price estimate and ETA, will be incorporated into the buyer confirmation message. For the MVP demonstration, this call will be made via mock service contracts.

- **WhatsApp Order Confirmation to Buyer**

  Once logistics is booked, the logistics agent via ILMU-GLM-5.1 will compose a natural language confirmation message and send it to the buyer via WhatsApp Business, including key messages such as confirmed items, quantities, total price, estimated delivery window, and a Lalamove or GrabExpress tracking link. The tone will be adapted based on the language and communication style, while adhering to the business logic.

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- **Wholesaler Staff Operations Dashboard**

  A web-based dashboard provides the wholesaler staff and operator to view all active and historical orders in real-time, showing the current pipeline state, buyer details, items and pricing. The staff can manually override any order details that are written to an audit log. Orders that require human attention such as low-confidence order parsing, API failures and timeouts will be surfaced via cards prominently.

- **Wholesaler Staff Alert Notification**

  For situations that require immediate staff attention as mentioned above, such as confirmation timeouts from buyers, logistical booking failures, low confidence parsing, the system will dispatch an alert to staff members via WhatsApp messages or email. Each alert contains the order ID, issue reasoning and editable fields in the card.

- **Order Log and Basic Reporting**

  All orders will be written to a persistent order log with audit trails, including timestamps of state transitions, inputs, parsed JSON, reasoning outputs and final delivery details. For basic reporting, daily order counts, value, fulfilment rate and substitution order rates are configurable to be viewed over a date range. The data will be stored in a structured database for future advanced analytics.

7. Features Not Included (Scope Control)

- **Telegram / SMS / Facebook Messenger / Other Channel Support**

  Despite the multi-agent architecture being channel-agnostic at the intake agent level, integrating additional messaging channels requires channel-specific API connectors and message format normalisation. This MVP is exclusively developed and considered on WhatsApp, which is the dominant channel for Malaysian SME wholesale operations. For post-MVP phases, other channels like Telegram and Facebook Messenger can be documented as planned extensions once validated on WhatsApp.

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- **Dynamic Multi-Drop Logistics Management**

  Orders that require multiple delivery addresses in a single trip requires multi-stop route optimisation, which is a distinct logistical issue from single-point deliveries. Although both Lalamove and GrabExpress can support multi-drop bookings, the business logic for consolidating orders into an efficient trip is planned for the post-MVP logistics module.

- **Customer ERP / Accounting System Synchronisation & Integration**

  Larger SME wholesalers may use their ERP or accounting software, which may vary significantly between platforms and each requires a bespoke connector. Integration schemas and methodology are huge gaps, thus this MVP uses Google Sheets as the inventory source of truth, and a webhook output for ERP consumption is planned for post-MVP.

- **In-App Payment Collection**

  Enabling buyers to pay directly within the WhatsApp conversation needs integration with a licensed payment gateway such as iPay88, compliance with Bank Negara Malaysia e-payment guidelines and merchant onboarding processes that extend beyond MVP timeline. For the MVP, the wholesaler will still invoice the buyer through their existing process, and payment integration can be a high-value post-MVP feature.

- **Driver Tracking & Live Map Feature**

  Real-time driver tracking is provided natively to buyers by Lalamove and GrabExpress via their tracking links. Building another duplicate tracking experience within SupplyLah would need high-frequency API polling, map rendering and driver location state management, which would be a large-scoped feature that is already solved by the logistics provider.

- **Full Analytics and Business Intelligence Dashboard**

  Beyond basic order log reporting that is included in the MVP, a production Business

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Intelligence layer would require product-level margin tracking, delivery performance benchmarking and GLM reasoning audit logs for compliance analysis. Building a data warehouse, pipeline and interactive charting layer is a massive engineering investment that is best for the post-MVP phase.

8. Assumptions & Constraints

- **LLM Cost Constraints:**

  For the audio input pathway, it would take 112,100 tokens for both speech-to-text and reasoning, whereas for the image input pathway, it would be around 3,950 tokens, and lastly for the text input pathway, it would be 2,050 tokens. To perform a blended average token usage per session, it would be around 40,000 tokens, where this large number is mostly driven by audio transcription, which is also low cost per million tokens.

  Our system utilises a multimodal approach to inputs, ingesting text, audio (voice notes and voice calls) and images of order list. For text inputs, the typical WhatsApp text can span from 30 to 100 words, which would account for 50 to 150 tokens. For audio, voice notes can last from 15 to 45 seconds, which ranges from 20,000 to 60,000 tokens. On the other hand, voice calls could take up to 3 minutes, which is around 240,000 tokens. For images, however, 1 to 2 would be sent per session, which would be around 2,000 tokens.

  Audio will be routed to Groq's OpenAI Whisper API, which would take an average duration of 83 seconds between voice notes (45 seconds) and voice calls (2 minutes), generating approximately 110,000 tokens. Next, the speech to text tokens that are passed through ILMU-GLM-5.1 API for structuring the order is about 150 tokens, then outputting 250 tokens of generated JSON. For images, processing a handwritten or printed list consumes about 2,000 tokens, through GLM-4.6V, and then to output a structured JSON of about 250 tokens. For pure text requests, an average of 100 input tokens is needed.

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Next, using ILMU-GLM-5.1 API for handling complex reasoning for the inventory and business logic agent, would take approximately 1,200 tokens (1,000 for input, 200 for output). By using Z.ai's dynamic context caching, we can reuse the wholesaler's system prompt, pricing, inventory schemas that remain static across all buyer sessions, which is about 800 tokens.

Then, the workflow goes to the third logistic agent, which uses the ILMU-GLM-5.1, processing logistics API response and generating the final WhatsApp confirmation message, consuming about 500 tokens (300 input, 200 output).

- **Technical Constraints:**
  - **Logistics Booking Limited to Lalamove Single-Drop (Mocked)**
    The logistics integration is scoped to single-drop Lalamove bookings only. Multi-drop routing, GrabExpress and other logistic providers are not included in the MVP. Besides, the Lalamove API call is made against a realistic mock service due to the live API credentials requiring Lalamove business verification that is not feasible within the hackathon scope. The mock will return structurally-sound responses to the live API.

  - **WhatsApp Channel Only in MVP**
    SupplyLah is connected to the WhatsApp Business API in the MVP, whereas other channels like Facebook Messenger, Telegram, SMS and WeChat are not supported. The system assumes that the operator will hold an approved WhatsApp Business account and that their buyers are reachable via WhatsApp. If the WhatsApp Business API approval is not completed, the team will fall back to a WhatsApp mimicking mock chat UI that drives the same agent backend pipeline and workflow.

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- **Performance Constraints:**
  - **Voice Call Transcription Adding 10 to 30 Seconds of Processing Lag**
    Live voice call transcription processes audio after the call ends rather than in real-time, introducing a lag of approximately 10 to 30 seconds before the transcribed order reaches the intake agent for parsing. Buyers are sent an acknowledgement message after finishing the call.

  - **End-to-end Order Processing Target Under 90 Seconds**
    For text and voice note orders, the system targets an end-to-end latency under 90 seconds from message receipt to the buyer receiving a pricing proposal. GLM inference latency is the primary variable. If response times exceed targets during peak load, requests are queued and buyers receive an acknowledgement within 10 seconds of their request message.

  - **Image Parsing Accuracy Dependent on Photo Quality**
    Handwritten order image parsing via GLM vision is constrained by the quality of the photograph submitted by the buyer. If the images are blurred, poorly lit, and contain heavily stylised handwriting, it may produce low-confidence parses. By applying a confidence threshold, parses below it will trigger a clarification request to the buyer instead of proceeding with the potentially incorrect order. For the MVP, there will be no automated image enhancement pipeline.

- **User Input Constraints:**
  - **System Requires Human Approval for Low-Confidence Orders**
    SupplyLah is based on the multi-agent workflows, but it is not without any human oversight in the MVP. Any order with low GLM confidence score will be escalated to the staff dashboard for manual review, prioritising order accuracy over speed for edge cases (such as blurry images and noisy audio environments)

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and ensuring the system is safe to be deployed in real wholesale operations without risking incorrect fulfilment.

- **GLM Cannot Initiate Order Without a Buyer Message Trigger**
  The system is based on reactive mode, which only processes an order when the buyer sends a message to the WhatsApp number. It does not proactively reach out to buyers for soliciting orders, generate demand forecasts autonomously or initiating outbound communication. All agentic reasoning is triggered by an inbound buyer message, together with confirmation of orders and follow-up nudging.

9. Risks & Questions Throughout Development

- **GLM Parsing Accuracy**
  Q: **Will GLM reliably parse the full range of Malaysian order messages?**
  A: Bahasa Rojak is highly contextual and varies significantly by region, industry and relationship between the buyer and seller. The risk is that GLM produces plausible but incorrect parses, for example, misidentifying a product name or quantity, that pass the confidence threshold and proceed to fulfilment without staff review.
  Mitigation: The confidence scoring system is tuned conservatively during testing against a dataset of real wholesale order messages, the staff override capability provides a catch-all fallback and GLM clarification messages are designed to catch ambiguity before the order is committed.

- **Voice Call Infrastructure**
  Q: **Can live call transcription be reliably built within the hackathon timeline?**
  A: Intercepting live WhatsApp voice calls programmatically requires a telephony relay layer (Twilio Voice or similar) that is architecturally more complex than the REST-based WhatsApp Business API. The risk is that this feature is not stable enough for a live demo.

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Mitigation: Voice note transcription is a fully functional fallback that covers the majority of voice-based ordering behaviour. If live call transcription is not demo-ready, it is presented as a near-term roadmap feature with the architecture documented and the infrastructure scaffolded.

- **Inventory Consistency**

  Q: **How do we prevent overselling when multiple orders are confirmed simultaneously?**

  A: If two buyers confirm orders for the same product at the exact same millisecond, a "Race Condition" could occur where both processes read the same stock level before either has finished deducting it.

  The system mitigates this natively using Supabase PostgreSQL Atomic Transactions. Instead of a slow, single-file serial queue, we implement a PL/pgSQL function (deduct_stock) called via Remote Procedure Call (RPC). This mechanism ensures:

  1. Row-Level Locking: When an update is initiated, PostgreSQL automatically places a lock on that specific product row. Any concurrent request for the same product must wait until the first transaction is committed or rolled back.
  2. Atomic Check-and-Deduct: The logic WHERE stock_quantity >= p_qty is executed inside the database kernel. This means the system check and the actual subtraction happen as a single, indivisible unit of work.
  3. Deterministic Success/Failure: The function returns a Boolean value. If the deduction is successful, it returns true. If stock is insufficient, it returns false, which immediately triggers the AI Inventory Agent to inform the buyer and propose a substitution or discount.

  This approach significantly increases system throughput compared to a serial queue, as it allows parallel processing of orders for different products while maintaining absolute mathematical integrity for every individual SKU.

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- **GLM Latency During Demonstration**

  Q: **Will GLM response times be acceptable during a live demonstration?**

  A: Live demos are sensitive to inference latency. A 20-second wait for a GLM response during a judge presentation creates a poor impression. The risk is that ILMU GLM's API latency is higher than expected under load.

  Mitigation: The team pre-tests latency under representative load conditions before the demo. A "processing" acknowledgement message is sent to the buyer within 10 seconds of any inbound message to signal activity and the demo script is designed to include a brief narrative explanation during the processing window so that latency reads as expected behaviour rather than a failure.

- **Image Parsing Reliability**

  Q: **How do we handle misreading a handwritten order image?**

  A: Handwritten text parsing via GLM vision introduces a higher error rate than text-based input, particularly for product shorthand and quantities that are ambiguous when handwritten. The risk is that an incorrect parse leads to a wrong order being fulfilled.

  Mitigation: The confidence gate ensures low-confidence image parses are escalated to staff review rather than proceeding automatically. GLM is prompted to extract only what it can read with certainty and to flag partial or uncertain fields explicitly and the buyer confirmation step acts as a final check, where buyers can reject a proposal that contains errors before the order is committed.

- **Scope Creep**

  Q: **How do we ensure the team stays focused on core agent logic during the hackathon?**

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A: With multiple interesting sub-problems, such as voice calls, image OCR, logistics APIs, a dashboard, there is a real risk that the team spreads effort too thin and delivers several half-finished features rather than a solid end-to-end workflow.

Mitigation: The team designates a desirable demo scenario (one text order, one voice note order) that must be fully functional before any stretch features are touched. The staff dashboard is built with a minimal viable UI that shows order state correctly, with visual polish as a secondary priority and all mocked components (Lalamove, WhatsApp fallback) are built first so the pipeline can be demonstrated end-to-end even if individual integrations are not live.

- **Data Privacy**

  Q: **How are buyer order messages and personal data handled?**

  A: Buyer WhatsApp messages contain personal data (name, address, ordering patterns) that is subject to Malaysia's Personal Data Protection Act (PDPA) 2010. The risk is that storing raw messages and transcripts without appropriate controls creates a compliance exposure.

  Mitigation: For MVP, all order data is stored in a single-tenant database controlled by the operator. Raw message content is retained only for the duration required to complete the order and is flagged for deletion after 90 days. The system does not share buyer data with any third party beyond the logistics API (which receives only the delivery address).

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